

Compact magnetometer project update

- The lock-in module now runs the full workflow:
ODMR sweep $\rightarrow$ 16 parked frequencies
 $\rightarrow \Delta f(t) \rightarrow \Delta B(t)$.
- Latest parking run: 8 kept transitions, 16 parked frequencies, no rejected fitted centers.
- Live lock-in test: 460 batches in 39.4 s, about 11.7 Hz update rate.
- Magnetic-field reconstruction is implemented with Yizhou's vector algorithm: 460/460 samples rank-ok.
- Live B-field measurement works, but online reconstruction slows the effective update rate.

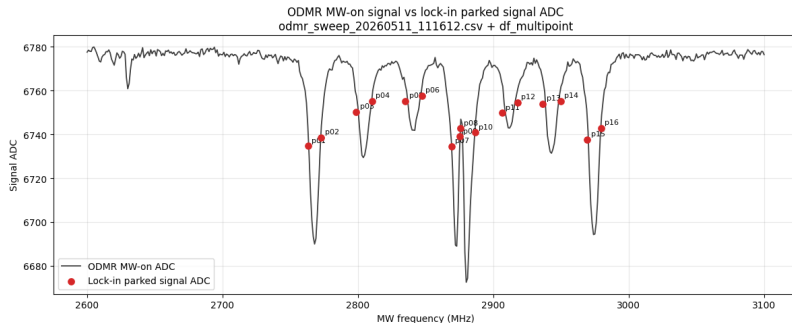
Item	Current result
Transitions	8
Parked points	16
Single batch	113 ms
Live lock-in	11.7 Hz
Live B-field	5.9 Hz

Workflow 1: choose 16 parked frequencies



- Start from the newest ODMR sweep.
- Fit 8 transition centers.
- Put two parked frequencies on the flanks of each transition.
- These flank points convert lock-in intensity changes into a frequency shift.

Workflow 2: acquire all parked points



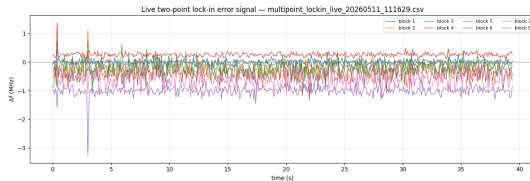
- One FPGA program measures all 16 frequencies.
- The output is one wide row per batch: signal, reference, and frequency for every parked point.
- Latest one-batch acquisition took 113 ms.

Workflow 3: convert lock-in signal to frequency shift

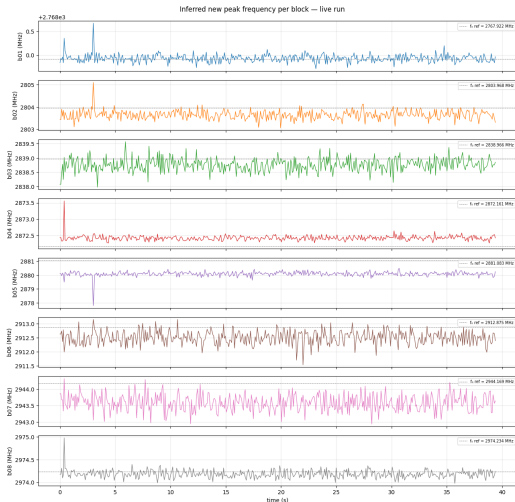
For each transition, use the two flanks:

$$\Delta f_k(t) = \frac{[S_+(t) - S_-(t)] - [S_{+,0} - S_{-,0}]}{m_- - m_+}.$$

- S_- and S_+ are the live signals at the lower and upper parked points.
- $S_{-,0}$ and $S_{+,0}$ come from the reference ODMR calibration.
- m_- and m_+ are the local ODMR slopes at those two parked points.
- This gives a frequency-shift time series for each of the 8 transitions.
- Latest live run: 460 batches over 39.4 s.
- Per-block frequency noise: 0.08 to 0.25 MHz standard deviation.

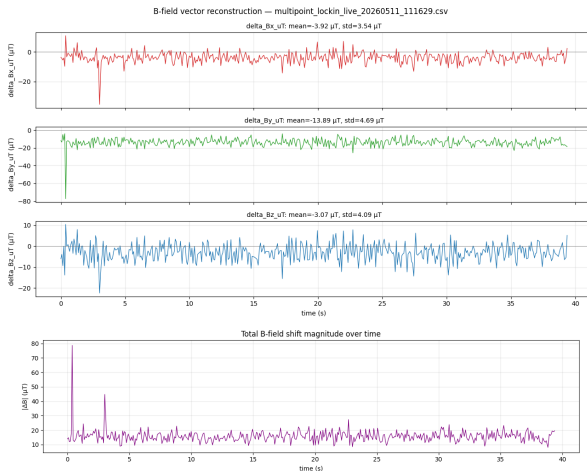


Workflow 4: inferred new peak frequencies



- This is the inferred new peak frequency for each of the 8 transitions.
- Solid traces show $f_{\text{new}}(t) = f_0 + \Delta f(t)$.
- Dashed lines mark the reference peak positions from the ODMR calibration.
- This view is useful for spotting sudden jumps before B-field reconstruction.

Workflow 5: reconstruct magnetic field

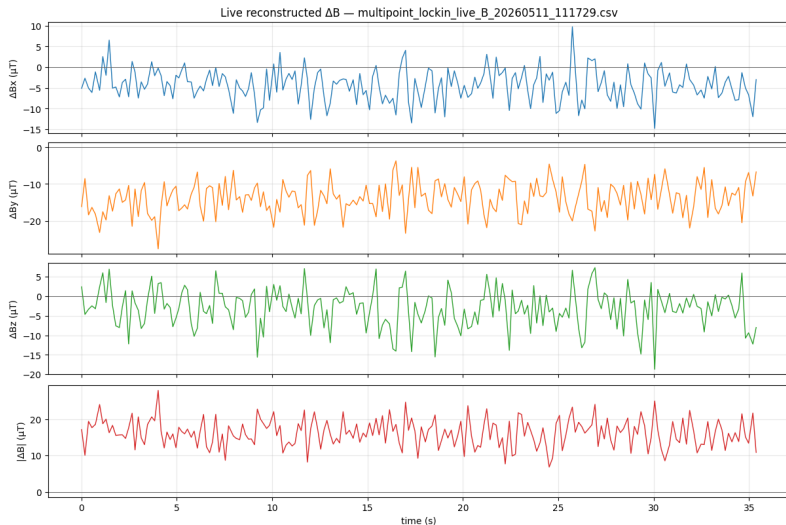


- 8 projected shifts to vector B-field.
- Offline live-run processing: 460 samples, 460 rank-ok.
- Projected-field noise: 2.9 to 8.9 μ T.
- Reconstructed B statistics:

Comp.	Mean	Std.
ΔB_x	-3.92	3.54
ΔB_y	-13.89	4.69
ΔB_z	-3.07	4.09
$ \Delta B $	15.79	4.43

Units: μ T

Workflow 6: live B-field measurement

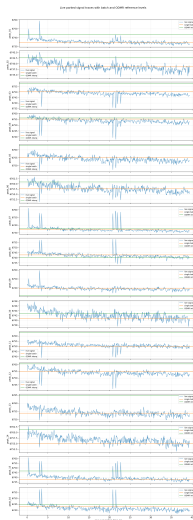
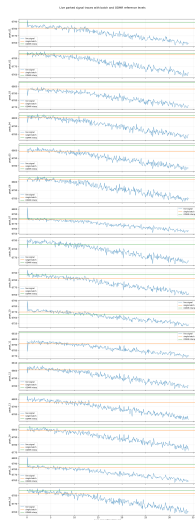


- Same reconstruction path as the offline workflow.
- Latest live B run: 208 batches in 35.4 s.
- Update rate: about 5.9 Hz.
- Main bottleneck: per-batch reconstruction and plotting.

Comp.	Mean	Std.
ΔB_x	-4.26	3.91
ΔB_y	-13.81	4.40
ΔB_z	-3.05	4.98
$ \Delta B $	16.22	3.80

Units: μT

Next 1: stabilize live traces and reject jumps



- Initial run: channels were relatively stable, but slowly decayed downward.
- Later runs: some parked channels developed sudden jumps.
- Short-term fix: reject sudden outliers before B reconstruction.
- Algorithm fix: add stricter checks so parked points avoid bad ODMR regions.
- Hardware diagnosis: separate warm-up/baseline drift from real peak jumps.

Next 2: verify reconstruction and speed up live B

Verify B reconstruction

- Apply a known test field and compare the recovered direction and magnitude.
- Compare offline reconstruction and live B reconstruction on the same saved lock-in data.
- Track residual norm, matrix rank, and condition number for every batch.

Make live B faster

- Decouple acquisition from reconstruction so data collection does not wait on plotting.
- Vectorize the per-batch calculation where possible.
- Update plots more slowly than the acquisition loop.
- Target: move live B closer to the raw live lock-in rate.

Next 3: system integration

- Dual-channel measurement: connect the lock-in workflow to the dual-channel PL readout.
- Hardware optimization: stabilize warm-up behavior, ADC baseline, microwave timing, and optical alignment.
- Data organization: keep the GitHub workflow clean so Yizhou's reconstruction code and the live data stay synchronized.
- Patent manuscript: document the 16-frequency parked lock-in workflow, live peak tracking, and vector B-field reconstruction.